OR4023 (Optimization) Homework 5

Name: _____ Student ID: _____

Question 1. Prove that a matrix $\mathbf{C}$ is symmetric and of rank-one if and only if $\mathbf{C} = \gamma w w^T$ for some scalar γ and unit vector w.

Question 2. Apply the symmetric rank-one quasi-Newton method to solve

$$\min\ f(x) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$$

with

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 & 1 \\ 2 & 7 & 3 \\ 1 & 3 & 9 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -9 \\ 0 \\ -8 \end{pmatrix}.$$

Initialize the method with $\mathbf{x}_0 = (0,0,0)^T$ and $\mathbf{B}_0 = \mathbf{I}$. Use an exact line search.

Question 3. Apply the BFGS quasi-Newton method to solve

$$\min f(x) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$$

with

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 & 1 \\ 2 & 7 & 3 \\ 1 & 3 & 9 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -9 \\ 0 \\ -8 \end{pmatrix}.$$

Initialize the method with $\mathbf{x}_0 = (0,0,0)^T$ and $\mathbf{B}_0 = \mathbf{I}$. Use an exact line search.

Question 4. Let $\mathbf{B}_{k+1}$ be obtained from $\mathbf{B}_k$ using the BFGS update formula. Prove that $\mathbf{B}_{k+1}\mathbf{s}_k = \mathbf{y}_k$.